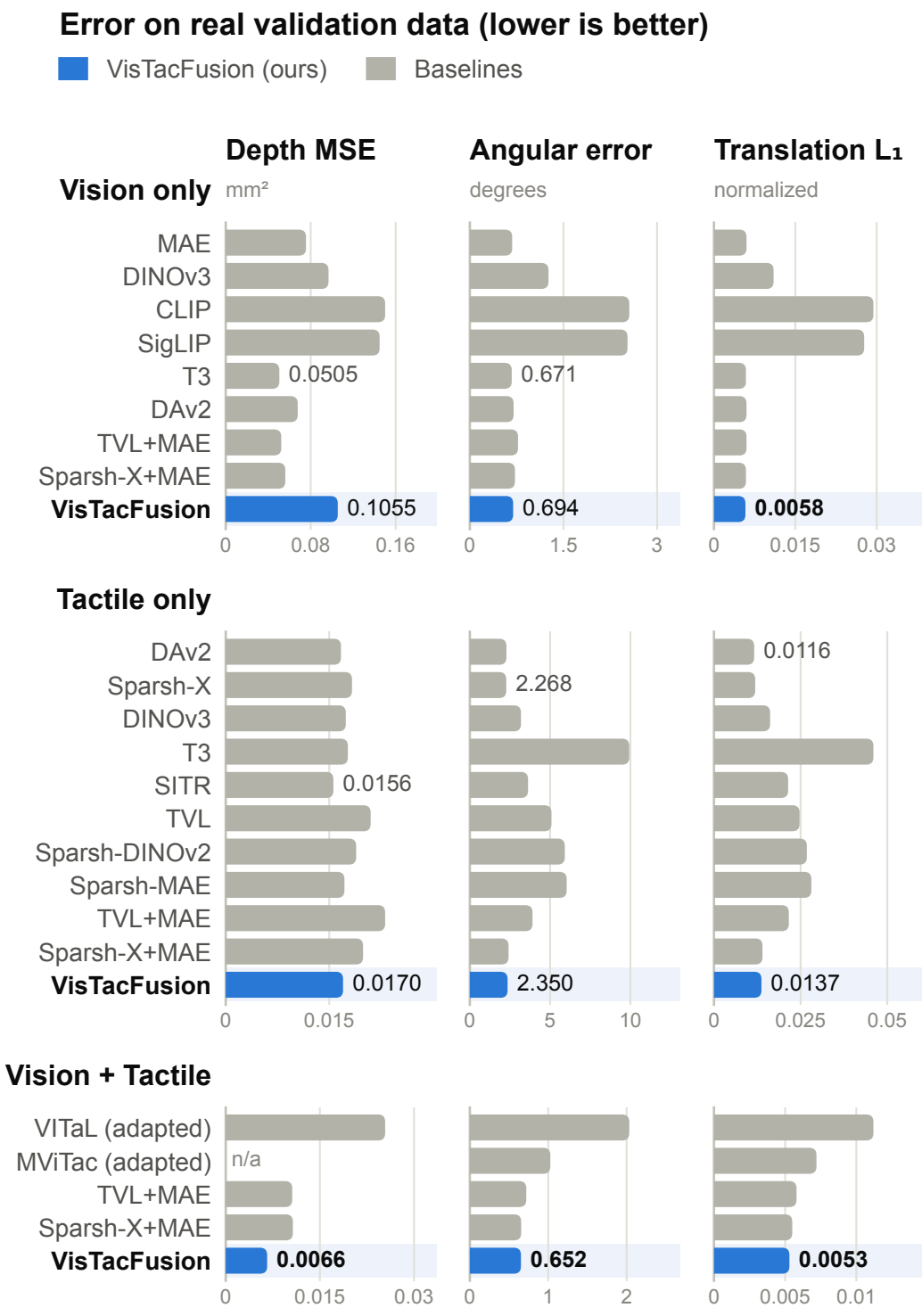
Error on real validation data (lower is better)
VisTacFusion (ours)
Baselines
Depth MSE
mm²
Angular error
degrees
Translation L₁
normalized
Vision only
MAE
DINOv3
CLIP
SigLIP
T3
DAv2
TVL+MAE
Sparsh-X+MAE
VisTacFusion
0.0505
0.671
0.1055
0.694
0.0058
0
0.08
0.16
0
1.5
3
0
0.015
0.03
Tactile only
DAv2
Sparsh-X
DINOv3
T3
SITR
TVL
Sparsh-DINOv2
Sparsh-MAE
TVL+MAE
Sparsh-X+MAE
VisTacFusion
0.0156
2.268
0.0116
0.0170
2.350
0.0137
0
0.015
0
5
10
0
0.025
0.05
Vision + Tactile
VITaL (adapted)
MViTac (adapted)
TVL+MAE
Sparsh-X+MAE
VisTacFusion
n/a
0.0066
0.652
0.0053
0
0.015
0.03
0
1
2
0
0.005
0.01